

Mike Roberts

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RESEARCH INTERESTS

Using **photorealistic simulation** for **embodied AI** and **computer vision**; motion planning, trajectory optimization, and control methods for robotics; reconstructing 3D scenes from images; continuous and discrete optimization; software tools and algorithms for creativity support

EDUCATION

Stanford University Stanford, California
Ph.D. Computer Science 2012–2019
Advisor: Pat Hanrahan
Dissertation: Trajectory Optimization Methods for Drone Cameras

Harvard University Cambridge, Massachusetts
Visiting Research Fellow Summer 2013
Advisor: Hanspeter Pfister

University of Calgary Calgary, Canada
M.S. Computer Science 2010

University of Calgary Calgary, Canada
B.S. Computer Science 2007

EMPLOYMENT

Adobe Research San Francisco, California
Senior Research Scientist 2024–
Mentor: Kalyan Sunkavalli

Intel Labs Seattle, Washington
Research Scientist 2021–2024
Mentor: Vladlen Koltun

Apple Seattle, Washington
Research Scientist 2018–2021
Mentor: Josh Susskind

Microsoft Research Redmond, Washington
Research Intern Summer 2016, Summer 2017
Mentors: Neel Joshi, Sudipta Sinha

Skydio Redwood City, California
Research Intern Spring 2016
Mentors: Adam Bry, Frank Dellaert

Udacity Mountain View, California
Course Developer, Introduction to Parallel Computing 2012–2013
Instructors: John Owens, David Luebke

Harvard University Cambridge, Massachusetts
Research Fellow 2010–2012
Advisor: Hanspeter Pfister

NVIDIA Austin, Texas
Developer Tools Programmer Intern Summer 2009

Radical Entertainment Vancouver, Canada
Graphics Programmer Intern 2005–2006

HONORS & AWARDS

Spotlight oral presentation, ECCV 2026 2026

Featured in the Highlights of SIGGRAPH session at the FMX Festival 2017 2017
1% selection rate (3 / 467)

Invited speaker, TEDxBerkeley 2017 2017

Excellent reviewer, CHI 2017 2017

Featured in the SIGGRAPH 2016 Technical Papers Trailer <i>4% selection rate (19 / 467)</i>	2016
Featured in the SIGGRAPH Asia 2015 Technical Papers Trailer <i>4% selection rate (11 / 302)</i>	2015
Front cover article, Cell 162(3)	2015
NSERC Alexander Graham Bell Canada Graduate Scholarship <i>14% selection rate (233 / 1628)</i>	2012

SOFTWARE & DATASETS

SPEAR: A Simulator for Photorealistic Embodied AI Research
github.com/spear-sim/spear

Hypersim: A Photorealistic Synthetic Dataset for Holistic Indoor Scene Understanding
github.com/apple/ml-hypersim

SELECTED PUBLICATIONS

My publications are also listed on [Google Scholar](https://scholar.google.com/citations?user=8888888888888888).

SPEAR: A Simulator for Photorealistic Embodied AI Research

Mike Roberts, Renhan Wang, Rushikesh Zavar, Rachith Dey-Prakash, Quentin Leboutet, Stephan R. Richter, Matthias Müller, German Ros, Rui Tang, Stefan Leutenegger, Yannick Hold-Geoffroy, Kalyan Sunkavalli, Vladlen Koltun
ECCV 2026

[*Spotlight oral presentation*](#)

PISA Experiments: Exploring Physics Post-Training for Video Diffusion Models by Watching Stuff Drop

Chenyu Li, Oscar Michel, Xichen Pan, Sainan Liu, **Mike Roberts**, Saining Xie
ICML 2025

Hypersim: A Photorealistic Synthetic Dataset for Holistic Indoor Scene Understanding

Mike Roberts, Jason Ramapuram, Anurag Ranjan, Atulit Kumar, Miguel Angel Bautista, Nathan Paczan, Russ Webb, Joshua M. Susskind
ICCV 2021

Submodular Trajectory Optimization for Aerial 3D Scanning

Mike Roberts, Debadepta Dey, Anh Truong, Sudipta Sinha, Shital Shah, Ashish Kapoor, Pat Hanrahan, Neel Joshi
ICCV 2017

Generating Dynamically Feasible Trajectories for Quadrotor Cameras

Mike Roberts, Pat Hanrahan
SIGGRAPH 2016

[*Featured in the Highlights of SIGGRAPH session at the FMX Festival 2017*](#)

[*Featured in the SIGGRAPH 2016 Technical Papers Trailer*](#)

An Interactive Tool for Designing Quadrotor Camera Shots

Niels Joubert*, **Mike Roberts***, Anh Truong, Floraine Berthouzoz, Pat Hanrahan
SIGGRAPH Asia 2015, * Authors contributed equally

[*Featured in the SIGGRAPH Asia 2015 Technical Papers Trailer*](#)

Saturated Reconstruction of a Volume of Neocortex

Narayanan Kasthuri, Kenneth Jeffrey Hayworth, Daniel Raimund Berger, Richard Lee Schalek, Jose Angel Conchello, Seymour Knowles-Barley, Dongil Lee, Amelio Vazquez-Reina, Verena Kaynig, Thouis Raymond Jones, **Mike Roberts**, Josh Lyskowski Morgan, Juan Carlos Tapia, H. Sebastian Seung, William Gray Roncal, Joshua Tzvi Vogelstein, Randal Burns, Daniel Lewis Sussman, Carey Eldin Priebe, Hanspeter Pfister, Jeff William Lichtman
Cell 162(3), 2015

[*Front cover article*](#)

Large-Scale Automatic Reconstruction of Neuronal Processes from Electron Microscopy Images

Verena Kaynig, Amelio Vazquez-Reina, Seymour Knowles-Barley, **Mike Roberts**, Thouis R. Jones, Narayanan Kasthuri, Eric Miller, Jeff Lichtman, Hanspeter Pfister
Medical Image Analysis 22(1), 2015

Design and Evaluation of Interactive Proofreading Tools for Connectomics

Daniel Haehn, Seymour Knowles-Barley, **Mike Roberts**, Johanna Beyer, Narayanan Kasthuri, Jeff W. Lichtman, Hanspeter Pfister
Transactions on Visualization and Computer Graphics 20(12), 2014

Neural Process Reconstruction from Sparse User Scribbles

Mike Roberts, Won-Ki Jeong, Amelio Vazquez-Reina, Markus Unger, Horst Bischof, Jeff Lichtman, Hanspeter Pfister
Medical Image Computing and Computer Assisted Intervention (MICCAI) 2011

A Work-Efficient GPU Algorithm for Level Set Segmentation

Mike Roberts, Jeff Packer, Mario Costa Sousa, Joseph Ross Mitchell
High Performance Graphics 2010

**INVITED
TALKS****SPEAR: A Simulator for Photorealistic Embodied AI Research**

3D in the Era of World Models, ECCV 2026 Workshop
Physical AI: Understanding and Building the Physical World, ECCV 2026 Workshop

September 2026

Sample-Efficient Learning with Synthetic Data

Imperial College London
University College London
Boston Dynamics AI Institute
Adobe Research
Manycore Tech
Toyota Research Institute
Facebook AI Research
Stanford University
Intel Labs
University of Washington

February 2025

August 2024

July 2024

September 2023

April 2021

January 2021

October 2020

Trajectory Optimization Methods for Drone Cameras

Oculus Research
Snapchat Research
Carnegie Mellon University
Boston University
Google Research
Adobe Research
Toyota Technological Institute at Chicago
NVIDIA Research
Simon Fraser University

June 2018

May 2018

March 2018

February 2018

Harnessing the Creative Power of Drones

Charles University in Prague
Google
University College London
Disney Research
ETH Zurich
University of Oxford
Max Planck Institute for Informatics
UC Berkeley
Samsung
TEDxBerkeley 2017
Autel Robotics
3D Robotics
UC Berkeley

November 2017

June 2017

May 2017

April 2017

March 2017

February 2017

	Columbia University	November 2016
	Yale University	
	Princeton University	
	Brown University	
	Intel	October 2016
	Generating Dynamically Feasible Trajectories for Quadrotor Cameras	
	FMX Festival 2017, Highlights of SIGGRAPH session	May 2017
	Adobe Research	September 2016
	Apple	August 2016
	Massachusetts Institute of Technology	
	Skydio	February 2016
	3D Robotics	January 2016
TEACHING EXPERIENCE	Udacity	2013–2018
	Course Developer, Introduction to Parallel Programming	
	<i>Instructors: John Owens, David Luebke</i>	
	<i>Developed course materials in 2012–2013, over 80,000 students enrolled in 2013–2018</i>	
	Stanford University	Spring 2018
	Course Assistant, Convolutional Neural Networks for Visual Recognition	
	<i>Instructors: Fei-Fei Li, Justin Johnson, Serena Yeung</i>	
	Stanford University	Winter 2018
	Course Assistant, Mathematical Methods for Robotics, Vision, and Graphics	
	<i>Instructor: Doug James</i>	
	Massachusetts Institute of Technology	Summer 2016
	Guest Lecturer, Advances in Imaging	
	<i>Instructor: Ramesh Raskar</i>	
	Harvard University	Fall 2013
	Course Contributor, Data Science	
	<i>Instructors: Hanspeter Pfister, Joe Blitzstein</i>	
	<i>Contributed lecture notes to the initial offering of Harvard's Data Science course in Fall 2013</i>	
	Harvard University	Winter 2012
	Teaching Fellow, Visualization	
	<i>Instructor: Hanspeter Pfister</i>	
	Harvard University	Fall 2011
	Teaching Fellow, Computing Foundations for Computational Science	
	<i>Instructor: Hanspeter Pfister</i>	
	Harvard University	Winter 2011
	Teaching Fellow, Massively Parallel Computing	
	<i>Instructors: Hanspeter Pfister, Nicolas Pinto</i>	
	University of Calgary	Winter 2006, 2007, 2008
	Guest Lecturer, Video Game Programming	
REVIEWING EXPERIENCE	3DV, CHI, CVPR, Eurographics, High Performance Graphics, ICRA, NeurIPS, SIGGRAPH, SIGGRAPH Asia, Robotics and Automation Letters, Transactions on Graphics, Transactions on Visualization and Computer Graphics	
GAME CREDITS	Scarface: The World Is Yours (PC, Playstation 2, Wii, Xbox)	2006
	<i>Radical Entertainment, Sierra</i>	
PRESS COVERAGE	New App Lets Drone Pilots Customize Flight Path and Camera Movement Before Takeoff	
	<i>Digital Trends</i> (October 19th, 2015)	
	Researchers Create Software for Designing Pro Drone Shots in a Virtual World	

Petapixel (October 16th, 2015)

Interactive Drone App Lets You Capture Aerial Shots Like a Pro

Engadget (October 15th, 2015)

These Stunning Images Will Take You on a Journey Through the Brain

Huffington Post (August 4th, 2015)

3D Color Images of the Brain Reveal its Glorious Unseen Detail

Popular Science (July 31st, 2015)

3D Brain Map Reveals Connections Between Cells in Nano-Scale

The Guardian (July 30, 2015)

Crumb of Mouse Brain Reconstructed in Full Detail

Nature News (July 30, 2015)

A Voyage into the Brain

National Geographic (February 2014)

What Makes Us Human?

BBC Horizon (July 3rd, 2013)

In Pursuit of a Mind Map, Slice by Slice

The New York Times (December 27th, 2010)

REFERENCES

Pat Hanrahan

CANON USA Professor of Computer Science and Electrical Engineering, Stanford University

Vladlen Koltun

Distinguished Scientist, Apple

Hanspeter Pfister

An Wang Professor of Computer Science, Harvard University

Adam Finkelstein

Professor of Computer Science, Princeton University

John Owens

Child Family Professor of Engineering and Entrepreneurship, UC Davis

Sudipta Sinha

Principal Researcher, Microsoft Research

Josh Susskind

Machine Learning Scientist, Apple